Qualifying exam in Algebra. Fall 2022.

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Part I. Definitions and Theorems. (each problem 5 points)

1. State the universal property of localization of a commutative ring.

Solution: Let R be a commutative ring. Recall that multiplicative subset $S \subset R$ is a subset closed under multiplication and containing the identity element. A multiplicative subset S is proper if $0 \notin S$. Given a multiplicative subset $S \subset R$ we can construct a new ring $S^{-1}R$ and homomorphism $i_S : R \to S^{-1}R$.

Universal property: Let $S \subset R$ be a proper multiplicative subset and let $f : R \to R'$ be a homomorphism of commutative rings such that $f(s)$ is invertible for any $s \in S$. Then there exists a unique homomorphism $\tilde{f} : S^{-1}R \to R'$ such that $f = \tilde{f} \circ i_S$.

2. State the theorem on adjointness of $\otimes$ and Hom.

Solution: Let R and S be rings. Let V be $(R, S)-$bimodule, U be $S-$module, and W be $R-$module. Then $V \otimes_S U$ is naturally $R-$module and $\mathrm{Hom}_R(V, W)$ is naturally $S-$module.

Theorem There exists an isomorphism of abelian groups

$$\mathrm{Hom}_R(V \otimes_S U, W) \simeq \mathrm{Hom}_S(U, \mathrm{Hom}_R(V, W))$$

natural in U, W, and V.

Part II. True or false. Give brief justification. (each problem 8 points)

1. Let $F, G : \mathcal{A} \to \mathcal{B}$ be isomorphic functors from category $\mathcal{A}$ to category $\mathcal{B}$, and $f, g \in \mathrm{Hom}_{\mathcal{A}}(A, A')$. Then $F(f) = F(g)$ if and only if $G(f) = G(g)$.

Solution: TRUE. Let $\alpha : F(A) \to G(A)$ and $\beta : F(A') \to G(A')$ come from an isomorphism of F and G. Then by definition of natural transformation we have $G(h) \circ \alpha = \beta \circ F(h)$ for any $h \in \mathrm{Hom}_{\mathcal{A}}(A, A')$. Equivalently, $G(h) = \beta \circ F(h) \circ \alpha^{-1}$ and $F(h) = \beta^{-1} \circ G(h) \circ \alpha$ which gives the answer.

2. Let $F = \mathbb{Q}(\sqrt[3]{2})$. Then for any $n \in \mathbb{Z}_{\geq 1}$, the degree $[F(\sqrt[n]{1}) : F] = \phi(n)$ (here $\phi(n)$ is the Euler function).

Solution: TRUE. Since $[F : \mathbb{Q}] = 3$, the statement is equivalent to $[F(\sqrt[n]{1}) : \mathbb{Q}] = 3\phi(n)$. Now $F(\sqrt[n]{1}) = \mathbb{Q}(\sqrt[n]{1})(\sqrt[3]{2})$ and $[\mathbb{Q}(\sqrt[n]{1}) : \mathbb{Q}] = \phi(n)$. Thus we need to show that $[\mathbb{Q}(\sqrt[n]{1})(\sqrt[3]{2}) : \mathbb{Q}(\sqrt[n]{1})] = 3$ or, equivalently, that the polynomial $x^3 - 2$ is irreducible over $\mathbb{Q}(\sqrt[n]{1})$. Since this polynomial has degree 3, we need to show that it has no roots in $\mathbb{Q}(\sqrt[n]{1})$. This is clear since the extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not normal (since the polynomial $x^3 - 2$ has some non-real roots) and any subextension of $\mathbb{Q}(\sqrt[n]{1})/\mathbb{Q}$ is normal (since this extension is abelian).

3. The embeddding functor from finite abelian groups to all abelian groups admits a right adjoint.

Solution: FALSE. Assume that $F : \{$ Abelian groups$\} \to \{$ finite Abelian groups$\}$ is the right adjoint. This means that $\mathrm{Hom}(A, B) = \mathrm{Hom}(A, F(B))$ for any finite abelian group A and abelian group B. In particular $\mathrm{Hom}(A, B)$ is finite for any such groups. This is not the case: take $A = \mathbb{Z}/2\mathbb{Z}$ and $B =$ infinite direct sum of copies of $\mathbb{Z}/2\mathbb{Z}$.

4. Every group of order 2022 is solvable.

Solution: TRUE. We have prime factorization $2022 = 2 \cdot 3 \cdot 337$. Thus any group of order 2022 has a unique Sylow 337-subgroup (since number of such subgroups is congruent to 1 modulo 337 and is a divisor of 2022). Thus this subgroup is normal and cyclic (hence abelian). The quotient by this subgroup is a group of order 6, hence solvable (by classification of groups of order 6, or by direct argument with Sylow 3-subgroup). An extension of solvable group by solvable (e.g. abelian) group is solvable and we are done.

5. For a ring R, any finitely generated $R-$module has a simple quotient.

Solution: TRUE. We know that this is the case for modules generated by one element, i.e. cyclic: such module is of the form R/I for some (left) ideal $I \subset R$ and any ideal is contained in a maximal ideal. Now let us apply induction in the number of generators: assume that module M is generated by n generators and let m be one of the generators. If $M = Rm$ then M is cyclic and we are done. Otherwise M/Rm is generated by $n-1$ generators and has a simple quotient by induction assumption.

Part III. Longer problems. (each problem 10 points)

You have to state precisely any standard result that you are using.

1. Let P be a square matrix over $\mathbb{Q}$. Prove that $P^2 = P$ if and only if $\mathrm{rank}(P) = \mathrm{tr}(P)$ and $\mathrm{rank}(\mathrm{Id} - P) = \mathrm{tr}(\mathrm{Id} - P)$ (here Id is the identity matrix).

Solution: Assume that $P^2 = P$. Then P is diagonalizable with eigenvalues 0 and 1; the result follows since the rank and trace are invariant under similarity and the result is obvious for diagonal matrices.

Conversely, assume $\mathrm{rank}(P) = \mathrm{tr}(P)$ and $\mathrm{rank}(\mathrm{Id} - P) = \mathrm{tr}(\mathrm{Id} - P)$. Then $\mathrm{rank}(P) + \mathrm{rank}(\mathrm{Id} - P) = \mathrm{tr}(P) + \mathrm{tr}(\mathrm{Id} - P) = \mathrm{tr}(\mathrm{Id}) = n$ where P is of size $n \times n$. Equivalently,

$$(n - \mathrm{rank}(P)) + (n - \mathrm{rank}(\mathrm{Id} - P)) = n.$$

Observe that $n - \mathrm{rank}(P) = \dim(\mathrm{Ker}(P)) =$ dimension of 0-eigenspace of P, and $n - \mathrm{rank}(\mathrm{Id} - P) =$ dimension of 1-eigenspace of P. Clearly these two subspaces intersect by zero. Thus the column space is a direct sum of these two subspaces; equivalently P is diagonalizable with eigenvalues 0 and 1. Hence $P^2 = P$.

2. Prove that the ring $\mathbb{Z}[\sqrt{7}]$ is integrally closed.

Solution: We need to show that any element of $\alpha \in \mathbb{Q}(\sqrt{7})$ (field of fractions of $\mathbb{Z}[\sqrt{7}]$) integral over $\mathbb{Z}[\sqrt{7}]$ is an element of $\mathbb{Z}[\sqrt{7}]$. Note that such α is integral over $\mathbb{Z} \subset \mathbb{Z}[\sqrt{7}]$ since integral extension $\mathbb{Z}[\sqrt{7}][\alpha]/\mathbb{Z}[\sqrt{7}]$ of integral extension $\mathbb{Z}[\sqrt{7}]/\mathbb{Z}$ is integral. Thus $irr(\alpha, \mathbb{Q})$ should be monic with integral coefficients (since by Gauss' Lemma any monic divisor of monic polynomial with integral coefficients has integral coefficents). Now if $\alpha = a + b\sqrt{7}$ with $a, b \in \mathbb{Q}$, we have $irr(\alpha, \mathbb{Q}) = (x-a)^2 - 7b^2$ (we can assume $b \neq 0$ without loss of generality); thus $2a$ and $a^2 - 7b^2$ should be in $\mathbb{Z}$. Thus $a \in \frac{1}{2}\mathbb{Z}$. If $a \in \mathbb{Z}$, then $-7b^2 \in \mathbb{Z}$ hence $b \in \mathbb{Z}$ (since $b \in \mathbb{Q}$). If $a \notin \mathbb{Z}$ then $a^2 \in \frac{1}{4} + \mathbb{Z}$ (since a square of odd number is $\equiv 1 \pmod 4$). Thus $7b^2 \in \frac{1}{4} + \mathbb{Z}$, so $b \in \frac{1}{2}\mathbb{Z}$ and $b \notin \mathbb{Z}$. This is a contradiction since $\frac{7}{4} \notin \frac{1}{4} + \mathbb{Z}$. We proved that any element of $\mathbb{Q}(\sqrt{7})$ integral over $\mathbb{Z}$ is from $\mathbb{Z}[\sqrt{7}]$ and we are done.

3. Let χ and ϕ be some irreducible complex characters of a finite group G with $\chi(1) \neq \phi(1)$. Let $\chi\phi = \sum_i \psi_i$ be the decomposition of product character $\chi\phi$ into sum of (not necessarily distinct) irreducible characters ψ_i. Prove that none of the characters ψ_i is linear, i.e. $\psi_i(1) \neq 1$ for all i.

Solution: We can and will assume that $\chi(1) > \phi(1)$ (interchange the roles of χ and ϕ otherwise). Let U, V, W_i be the irreducible representations with characters χ, ϕ, ψ_i; by the assumption we have

$$U \otimes V \simeq \bigoplus_i W_i.$$

In particular, $\mathrm{Hom}(U \otimes V, W_i) \neq 0$. Hence $\mathrm{Hom}(U, W_i \otimes V^*) \neq 0$. Any nonzero homomorphism $U \to W_i \otimes V^*$ must be an embedding since U is irreducible; hence $\chi(1) = \dim U \leq \dim W_i \otimes V^* = \psi_i(1)\phi(1)$. If $\psi_i(1) = 1$ this contradicts to $\chi(1) > \phi(1)$.

4. Assume an algebraic set $\mathcal{V}$ is a disjoint union of two non-empty closed subsets $\mathcal{V}_1$ and $\mathcal{V}_2$. Prove that the coordinate algebra of $\mathcal{V}$ contains a nontrivial idempotent.

Solution: Let $I, I_1, I_2 \subset F[x_1, \ldots, x_n]$ be the (radical) ideals corresponding to $\mathcal{V}, \mathcal{V}_1, \mathcal{V}_2$. Then $I = I_1 \cap I_2$. We also have $I_1 + I_2 = F[x_1, \ldots, x_n]$ (since any maximal ideal containing $I_1 + I_2$ would give a point in the intersection of $\mathcal{V}_1$ and $\mathcal{V}_2$). Thus

$$F[\mathcal{V}] = F[x_1, \ldots, x_n]/I \simeq F[x_1, \ldots, x_n]/I_1 \oplus F[x_1, \ldots, x_n]/I_2$$

by the Chinese Remainders theorem. In particular, $F(\mathcal{V})$ contains a nontrivial idempotent corresponding to $(1, 0) \in F[x_1, \ldots, x_n]/I_1 \oplus F[x_1, \ldots, x_n]/I_2$.

5. Let A and B be two division algebras. Prove that A is Morita equivalent to B if and only if $A \simeq B$.

Solution: It is clear that $A \simeq B$ implies that A and B are Morita equivalent. Let us prove converse. Assume that A and B are Morita equivalent, i.e. there exists an equivalence $F : A\text{-}Mod \to B\text{-}Mod$. Recall that F is automatically additive functor. We know that any $A-$module is completely reducible and there is a unique up to isomorphism simple module, namely A itself. Thus $A \in A\text{-}Mod$ is a unique (up to isomorphism) object which does not decompose into a direct sum of other nonzero objects. Thus $F(A) \in B\text{-}Mod$ should have the same property since F (and its inverse) is additive. Hence $F(A) \simeq B$. Now $\mathrm{End}_A(A) = A$ and $\mathrm{End}_B(F(A)) = \mathrm{End}_B(B) = B$. Since F is an equivalence, it induces an isomorphism of rings $A = \mathrm{End}_A(A) \to \mathrm{End}_B(F(A)) = B$.